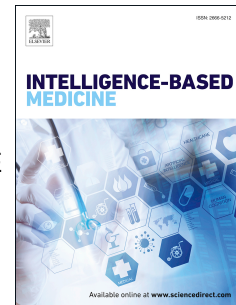


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STViTDA-Net: An Explainable Transformer-Based Framework with STGAN-ViT-MAE and Deformable Attention for Multi-Class Skin Cancer Classification

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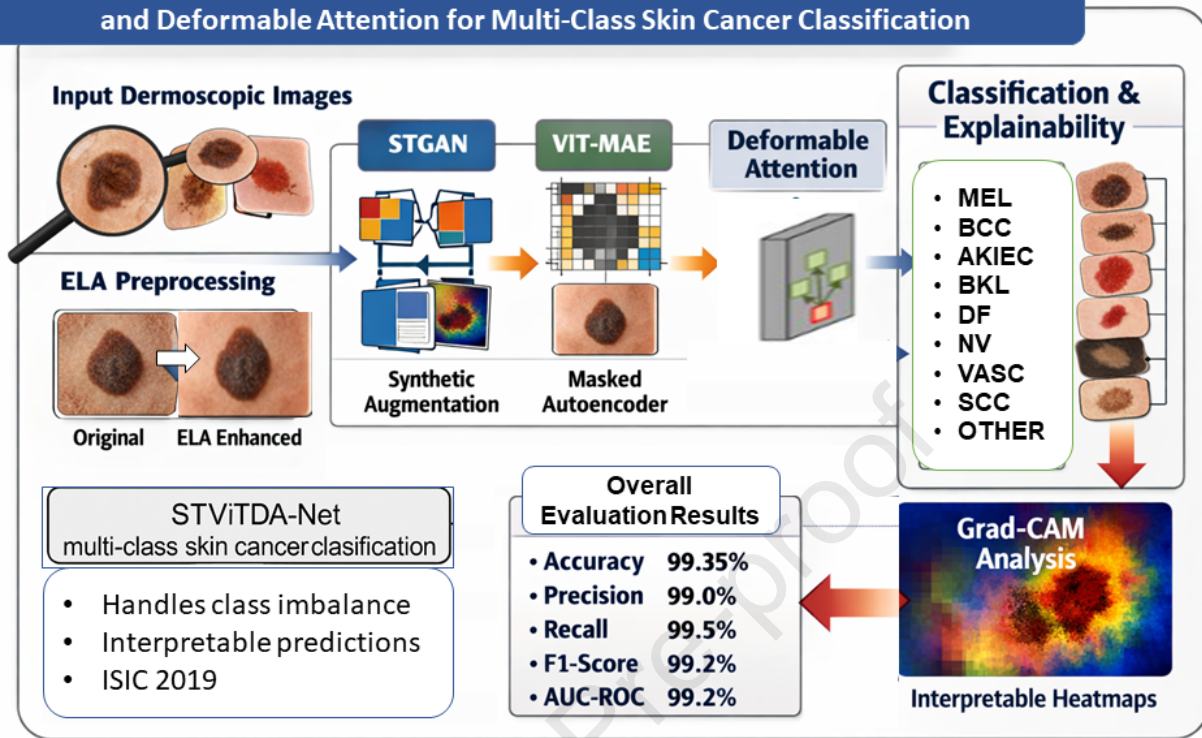
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STViTDA-Net: An Explainable Transformer-Based Framework with STGAN-ViT-MAE and Deformable Attention for Multi-Class Skin Cancer Classification

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ABSTRACT

Skin cancer continues to pose a major global health challenge, and its early identification is essential for improving patient outcomes. Traditional diagnostic practices rely heavily on clinician expertise and manual interpretation of dermoscopic images, making the process subjective, inconsistent, and time-consuming. To address these limitations, this work introduces STViTDA-Net, an explainable transformer-based framework designed for fast, objective, and scalable multi-class skin cancer classification. The model integrates three key components: STGAN for class-balanced dermoscopic image augmentation, ViT-MAE for robust hierarchical feature learning through masked patch reconstruction, and a Deformable Attention Transformer Encoder that adaptively focuses on irregular lesion boundaries and subtle spatial variations. Preprocessing with Error Level Analysis (ELA) enhances fine-grained diagnostic cues, while Grad-CAM provides interpretable heatmaps that highlight the regions influencing the model's predictions. Unlike manual dermoscopic evaluation, STViTDA-Net performs end-to-end inference within milliseconds and delivers consistent, expert-independent predictions supported by visual explanations. When evaluated on the ISIC2019 dataset comprising nine lesion categories, the model achieves 99.35% accuracy, 99.0% precision, 99.5% recall, 99.2% F1-score, and 99.2% AUC-ROC, surpassing existing CNN and transformer baselines. By unifying class-balanced augmentation, adaptive feature encoding, deformable attention, and explainable outputs, STViTDA-Net establishes a powerful and efficient solution for automated dermatological diagnosis.

Keywords: Skin Cancer Classification, STGAN, ViT-MAE, Deformable Attention, Grad-CAM, Explainable AI, ISIC2019 Dataset, Medical Image Augmentation

1. INTRODUCTION

Skin cancer turned out to be the most frequently diagnosed malignancy in the world; the portion of morbidity and mortality caused by the presence of skin cancer was reached by **melanoma and non-melanoma variants**. Another type of cancer numbers 2-3 million according to the World Health Organization, and there are also 132,000 cases of skin cancer called melanoma [1]. The key to successful diagnosis is early and accurate diagnosis, as the prognosis becomes dismal once the disease has progressed to an advanced stage. As an example, the five-year survival for melanoma in the early stages is greater than 99%, whereas metastatic cases have a survival rate of less than 30% [2]. Nevertheless, **effective early detection** remains a significant challenge, particularly in areas where people lack access to dermatologists and high-quality diagnostic devices [3]. Conventional diagnosis is highly dependent on the use of **dermoscopy and visual experience**, which are subjective and prone to **inter-observer variation** [4]. Such constraints underscore the need for intelligent, scalable, and explainable AI-based systems that can automate and assist in the early detection of skin cancer at expert levels. Moreover, the availability of large-scale datasets with dermoscopic images is growing, as seen in ISIC 2019. The current opportunity is to create **data-driven systems** that assist clinicians in decision-making and have the potential to detect subtle lesion characteristics [5]. However, existing automated diagnostic systems still face critical limitations, including poor handling of class imbalance, difficulty in capturing irregular lesion morphology, and limited ability to learn both global context and fine-grained visual cues in a unified manner. Moreover, most models lack clinically meaningful interpretability, which restricts their deployment in real-world dermatological settings. Therefore, there is a clear need for an integrated framework that simultaneously addresses **data imbalance, hierarchical feature representation, adaptive spatial discrimination**, and explainable **decision-making gaps** that the proposed STViTDA-Net is specifically designed to fill.

The classification of skin lesions through deep learning has become feasible due to the accuracy demonstrated by CNNs (e.g., ResNet, DenseNet), achieving expert-level performance on datasets such as HAM10000 and ISIC2019 [5][6]. Nevertheless, CNNs tend not to recognise the spatial context in the world, which is necessary for accurately identifying morphologically similar lesions. Transformer-based architectures, particularly Vision Transformers (ViT), have demonstrated potential due to their ability to **model long-range dependencies**, but large dataset sizes are relatively standard [7]. This is addressed in ViT-MAE through masked reconstruction, which is self-supervised to **encourage feature learning even with scarce data** [8]. Due to its ability to alleviate the problem of **class imbalance, GAN-based augmentation**, especially DCGAN and CycleGAN, has been employed, with the proposal of STGAN (Self-Transfer GAN) to enhance diversity in training through minority-class-aware generation [9]. The level of interpretability is achieved by using methods such as Grad-CAM, which allows for visual confirmation of the attention provided by the model [10]. Nevertheless, these developments tend to operate as silos, lacking a unified theory regarding generative augmentation, hierarchical learning of features, deformable attention priors, and **graphical** visualization.

Notably, the prevailing methods have a number of limitations that limit their performance and **clinical usability**. The CNNs poorly model global lesion structure and thus end up misclassifying the ambiguous cases [5][7]. Transformers correct this, but transformers need a large amount of data and can still overfit on unbalanced data [7][8]. GAN-based models produce **low-fidelity samples** that cannot accurately capture the morphology of lesions and tend to collapse on minor classes, such as DCGAN and CycleGAN, to name a few [9]. Although STGAN helps address the imbalance problem, it is not commonly combined with transformer model-based classifiers. Additionally, these popular interpretability techniques often produce **rough saliency maps** that are unlikely to align with clinically meaningful features. Besides, the

majority of current frameworks lack spatial adaption to lesion shape irregularities and do not provide an end-to-end solution.

Driven by the need to address data imbalance and other skin cancer classification problems using automated methods, as well as the limited contextual understanding and interpretability of clinicians, this paper proposes STViTDA-Net, a unified and interpretable framework based on the transformer architecture. This framework combines **generative augmentation, masked feature extraction, spatial adaptiveness, and visual explainability**. The model, in a synergistic manner, integrates Self-Transfer GAN (STGAN) [9] to generate synthetic minority-class lesion images and ViT-MAE (Vision Transformer with Masked Autoencoder) [8] as a durable and hierarchical feature extractor. Here, adapting to the regions of interest (lesion regions in this case) with arbitrary and irregular shapes, an attention transformer encoder (also a deformable attention encoder) will be used to learn the attention offsets in a data-driven manner [11]. The STGAN module produces **high-resolution, morphologically sound synthetic dermoscopic images** to counterbalance the underrepresented classes in the ISIC2019 dataset [11], thereby enhancing model generalizability and resistance to class inference. In the meantime, ViT-MAE, through masked image reconstruction, learns to store both contextual dependencies and fine-grained lesion features, even when semi-supervised constraints limit the framework.

To further enhance spatial modelling, Deformable Attention [11] is embedded within a transformer encoder, enabling the network to dynamically attend to irregular lesion boundaries and adaptive regions of interest. Preprocessing involves Error Level Analysis (ELA) [12], which accentuates compression artifacts and boundary anomalies to improve signal-to-noise discrimination. The classification pipeline concludes with a fully connected layer and employs **Grad-CAM [10] to generate saliency maps** that identify diagnostically significant lesion areas, thereby offering interpretability for clinical validation. Performance evaluation using the ISIC2019 dataset [13] validates the effectiveness of STViTDA-Net. The model achieves an accuracy of 99.35%, a precision of 99.0%, a recall of 99.5%, and an F1-score of 99.2%, alongside an AUC-ROC of 99.2%, surpassing conventional architectures, including ResNet50, standard ViT, and hybrid CNN-GAN combinations. This confirms the model's robustness across lesion categories and its capacity for reliable clinical-grade classification.

In summary, the proposed STViTDA-Net framework offers the following contributions:

- **Hybrid Model Design:** Combines STGAN for targeted lesion augmentation and ViT-MAE for self-supervised hierarchical feature encoding, enhancing classification accuracy on imbalanced datasets.
- **Adaptive Attention Mechanism:** Incorporates deformable attention to model irregular lesion geometries and improve spatial feature localization flexibly.
- **Enhanced Interpretability:** Employs ELA during preprocessing and Grad-CAM during inference to highlight salient lesion regions, providing clinical transparency and diagnostic insight.
- **Superior Performance:** Demonstrates state-of-the-art metrics on the ISIC2019 dataset: Accuracy: 99.35%, Precision: 99.0%, Recall: 99.5%, F1-Score: 99.2%, AUC-ROC: 99.2% outperforming prior CNN and transformer baselines in both robustness and explainability.

The arrangement of this paper is as follows: Section 2 provides a brief overview of the relevant literature and highlights important caveats; Section 3 describes the technique and system architecture of STViTDA-Net; Section 4 presents experimental results and comparison assessments; and Section 5 concludes with key findings and recommendations for future study.

2. RELATED WORKS

Research using deep learning techniques for the diagnosis of skin cancer is reviewed in this section. It addresses the progress of automated diagnostic systems, lesion categorization structures, and the strengths and weaknesses of current methodologies in reliably diagnosing skin cancers.

A recent study [14] proposed a hybrid **multi-class skin cancer classification framework** combining DeepLabV3+ for lesion segmentation and a triple-stream transformer architecture (Derm-ViT, Swin Transformer V2, and ConvNeXt V2) for robust feature extraction. Features were selected using ReliefF and classified via **k-nearest neighbours**, achieving "94.42% accuracy, 94.13% precision, 92.99% sensitivity, and 98.96% specificity" on the ISIC2019 dataset. This approach effectively captured **dermatological patterns and outperformed existing transformer-based models** in clinical relevance and diagnostic consistency.

A recent study [15] introduced Self-Transfer GAN (STGAN) to address the class imbalance in skin cancer datasets by generating high-quality 256×256 dermoscopic images through a two-stage learning process that captures both global and class-specific features. Evaluated on HAM10000, STGAN outperformed StyleGAN2 with improvements of 16% in FID and Inception Score, 4% in Precision, and 33% in Recall, leading to robust classification results "with 98.23% accuracy, 88.85% sensitivity, 90.23% precision, 89.48% F1-score, and 98.34%" specificity. This demonstrates STGAN's effectiveness in enhancing minority class representation and improving the performance of diagnostic models.

A recent study [16] proposed a multiscale attention-enhanced Vision Transformer (ViT) framework combined with CNNs to improve the differentiation between **benign and malignant skin lesions**. By leveraging attention maps to localise diagnostically relevant regions and integrating ensemble learning via majority voting, the ViT-EfficientNet hybrid model achieved an accuracy of 95.05% on the ISIC2018 dataset. This attention-based, ensemble-driven approach demonstrates improved robustness and accuracy in automated skin cancer classification.

In [17], a deep learning approach was developed that combined high-performing CNNs with soft attention mechanisms, achieving 93.4% accuracy on the HAM10000 dataset. Their method incorporated robust data preprocessing, including the removal of duplicates and class balancing through the use of **oversampling and undersampling**. The performance improvement was primarily attributed to the effective handling of class imbalance, highlighting the value of attention-enhanced CNN architectures for skin lesion classification.

The study by [18] introduced **AMViT-CNN, an attention-based multiscale Vision Transformer framework integrated with CNNs**, to address the challenges of fine-grained visual classification (FGVC), where subtle inter-class differences complicate discrimination. By incorporating a dual-path hierarchical pyramid top-down feature route and a bottom-up channel-spatial focus path, along with DropBlock, SENet, and global covariance pooling (GCP), the model enhances localisation and feature extraction. Experimental results on public FGVC datasets demonstrated the model's superior performance in capturing discriminative details critical for accurate classification.

The work presented in [19] proposed ViT-ALN, a Vision Transformer model enhanced with Adaptive Learning and Normalisation techniques, such as GELU activation, Layer Normalisation, and Dropout, tailored for skin cancer classification using the **HAM10000 dataset**. By training on rescaled and enhanced dermoscopic images, ViT-ALN achieved a classification accuracy of 96.8%, outperforming conventional CNN-based approaches. The model demonstrated improved performance across 250 training iterations, validating the effectiveness of its architectural refinements in dermatological diagnostics.

The research in [20] introduced a **Modified Multi-Head Attention Transformer (MMHAT)** architecture for **skin lesion classification**, specifically targeting malignant melanoma and melanocytic nevus. By leveraging self-attention, the model surpasses the limitations of traditional CNNs tied to kernel size and depth. Evaluated on the ISIC2018 dataset, MMHAT outperformed ResNet50 (90% accuracy) and Inception V3 (84% accuracy), with results confirming that multi-head attention significantly enhances classification accuracy in skin cancer diagnosis.

In [21], the authors proposed SClusterFormer, **Deformable convolution-enhanced multilevel Vision Transformer for hyperspectral image categorization**. To mitigate spectral-spatial information loss in standard ViT models[22], they introduced a spectral-spatial cluster attention mechanism leveraging spectral angle and Euclidean distances for refined local and global feature capture. The architecture integrates a **dual-branch multiscale** deformable convolution framework and a **cross-feature** fusion module, enabling collaborative feature learning. Experimental results on multiple HSIC datasets confirmed that SClusterFormer outperforms existing state-of-the-art methods.

The study in [23] proposed BiADATU-Net, a **high-precision segmentation model** for skin cancer images that combines deformable attention Transformers with bidirectional attention blocks in a **U-Net architecture**. The encoder captures both global and local features using dual attention, while the decoder employs scSE modules and deformable convolutions to model irregular lesion structures. Evaluated on the ISIC2016, ISIC2017, ISIC2018, and PH2 datasets, the model achieved an accuracy of **over 96% and demonstrated strong potential for clinical application.**

In [24], the authors proposed SkinTrans, an enhanced ViT-based model for skin cancer classification, leveraging self-attention to highlight relevant features and suppress noise. The method incorporates Contrastive learning, multiscale patch embedding, and overlapping sliding windows to increase feature representation and class separation. Evaluated on HAM10000 and a clinical dermoscopy dataset, SkinTrans achieved 94.3% and 94.1% accuracy, respectively, demonstrating its effectiveness in handling diverse dermoscopic images.

The study in [25] introduced a transformer-based model for segmenting **histopathological skin images into inflammation, malignancies** (BCC, SCC, IEC), and relevant tissue borders, aiming to aid in the classification of non-melanoma skin cancer. Evaluated on public datasets, it achieved an F1-score of 0.908, mIoU of 0.653, and 83.1% accuracy, with uncertainty maps enhancing interpretability. This framework supports pathologists in decision-making and advances toward fully automated diagnostic systems.

The study in [26] proposed a **hybrid ensemble deep learning model for skin cancer classification and segmentation on non-uniform datasets**, utilising Residual Learning, Swin Transformers, and Feed-Forward Neural Networks (FNN). Evaluated on ISIC-2008, PH-2, and HM007, the model achieved 98.78% accuracy, 98.7% precision, 98.7% F1-score, 98.64% recall, and 0.9863 MCC, outperforming existing methods. This robust framework represents a significant advancement in the accurate and consistent diagnosis of skin cancer.

The study in [27] proposed a **hybrid model that combines skin cancer classification feature extraction**, improved by ConvNeXtV2 blocks and separate self-attention. Trained on the ISIC2019 dataset, the model achieved 93.48% accuracy, 93.24% precision, 90.70% recall, and 91.82% F1-score, outperforming 20 CNN and ViT-based models. With only 21.92M parameters, it offers high accuracy and efficiency suitable for clinical deployment.

Recent advancements highlight the effectiveness of hybrid deep learning and explainable AI across diverse medical imaging tasks. PediaPulmoDx[28] combines CLAHE, Otsu

thresholding, LBP, HOG, DenseNet121, and XAI techniques to achieve 99.60% sensitivity, 99.80% specificity, 99.70% F1, 99.97% AUC, and 99.97% accuracy for pediatric pneumonia detection. CMV2U-Net[29] introduces an edge-weighted U-Net with dual-stream extraction, hierarchical fusion, and channel attention, attaining high AUC and F1-scores and demonstrating strong robustness against noise and compression artefacts. A 3D ResNet50 model[30] trained on ISIC 2024 3D-TBP achieves 93.88% accuracy (vs. 85.67% with 2D data), underscoring the diagnostic value of 3D lesion representation. An optimization-driven breast cancer framework[31] integrates adaptive thresholds across multiple optimizers and achieves 99.72% accuracy and 100% sensitivity, offering a highly reliable solution for early detection.

In the domain of dermatological diagnosis, [32] presented an explainable AI-driven approach for skin cancer detection by incorporating deep learning models with visual-based interpretability techniques. The study utilised two pre-trained Vision Transformers (ViT-base and ViT-large), three Swin Transformers, and five pre-trained CNN models, evaluating them across multiple metrics. To enhance class balance, SMOTE was employed during training on publicly available datasets. Explainability was achieved using Grad-CAM, Grad-CAM++, and Score-CAM, offering transparency in model decision-making. Among all architectures, ResNet50 achieved the best performance, with 88.8% accuracy, 86.9% precision, 88.6% sensitivity, 87.8% F1-score, and 88.9% specificity. XAI visualisations confirmed that the model focused on clinically relevant regions for lesion classification. Table 1 presents recent literature on skin cancer disease classification.

Table 1: Recent literature on skin cancer disease classification using deep learning

Ref No.	Methodology	Dataset	Measures	Research Remarks
[14]	DeepLabV3+ + Triple-stream Transformer (Derm-ViT, Swin V2, ConvNeXt V2) + ReliefF + KNN	ISIC2019	Acc: 94.42%, Prec: 94.13%, Sens: 92.99%, Spec: 98.96%	Transformer fusion effectiveness and real-time generalization not explored
[15]	Self-Transfer GAN (STGAN) for data generation	HAM10000	Acc: 98.23%, Sens: 88.85%, Prec: 90.23%, F1: 89.48%, Spec: 98.34%	Still lacks testing on multimodal clinical scenarios
[16]	Multi-scale Attention ViT + CNN + Ensemble Voting	ISIC2018	Acc: 95.05%	Limited validation in multi-class imbalanced datasets
[17]	CNN + Soft Attention + Preprocessing	HAM10000	Acc: 93.4%	Generalizability across datasets not proven
[18]	AMViT-CNN with multi-scale attention + DropBlock + SENet + GCP	FGVC datasets	Qualitative results, not reported in exact numbers	Not tested on skin lesion datasets
[19]	ViT-ALN (Adaptive Learning & Normalization)	HAM10000	Acc: 96.8%	Comparative complexity vs. traditional ViTs not reported
[20]	Modified Multi-Head Attention Transformer (MMHAT)	ISIC2018	Acc: >90% (better than ResNet50 and InceptionV3)	Focus on only two classes: melanoma and nevus

[21]	SClusterFormer with cluster attention + deformable conv	HSIC	Outperformed SOTA	Focus not on dermoscopic or clinical skin lesion datasets
[22]	ViT-based classification vs. 11 CNNs	Custom + HAM10000	Acc: 92.14%	Lack of interpretability and lesion-level analysis
[23]	BiADATU-Net (Deformable Attention Transformer U-Net)	ISIC2016, 2017, 2018, PH2	Acc: >96%	Real-time segmentation feasibility unaddressed
[24]	SkinTrans (Enhanced ViT + Contrastive Learning)	HAM10000, Clinical	Acc: 94.3%, 94.1%	No segmentation capability included
[25]	Transformer for histopathology + Uncertainty maps	Public Histopathology Datasets	F1: 0.908, mIoU: 0.653, Acc: 83.1%	Scope limited to non-melanoma categories
[26]	Hybrid: Residual Learning + Swin Transformer + FNN	ISIC-2008, PH2, HM007	Acc: 98.78%, Prec: 98.7%, F1: 98.7%, MCC: 0.9863	Explainability tools not included
[27]	ConvNeXtV2 + Separable Self-Attention	ISIC2019	Acc: 93.48%, Prec: 93.24%, Recall: 90.70%, F1: 91.82%	No use of interpretable models like Grad-CAM
[28]	PediaPulmoDx: Advanced preprocessing (CLAHE, Otsu), LBP + HOG feature extraction, DenseNet121, Grad-CAM	Pediatric Chest X-rays	Sens: 99.60%, Spec: 99.80%, F1: 99.70%, AUC: 99.97%, Acc: 99.97%	Strong preprocessing + XAI pipeline;
[29]	CMV2U-Net: Edge-weighted U-Net with dual-stream feature extractor, hierarchical fusion, channel attention	Multiple public image-splicing datasets	High AUC, High F1-scores (specific values reported per dataset)	Outperforms SOTA in forgery localization; robust to noise
[30]	3D ResNet50 with transfer learning + targeted augmentation	ISIC 2024 3D-TBP	Acc: 93.88% (3D), 85.67% (2D)	Demonstrates superiority of 3D data for multi-class skin cancer
[31]	Optimization-driven neural network with adaptive thresholds optimizers	Wisconsin Breast Cancer Dataset	Acc: 99.72%, Sens: 100%	Momentum-threshold early stopping improves convergence and reduces overfitting;
[32]	Explainable ViTs & CNNs + Grad-CAM++/Score-CAM + SMOTE	Public datasets	Best: Acc: 88.8%, Prec: 86.9%, Sens: 88.6%, F1: 87.8%, Spec: 88.9%	Performance not best, but highest interpretability

Despite significant advancements in deep learning for skin cancer classification, existing models face critical limitations in handling class imbalance, capturing irregular lesion patterns, and providing clinical interpretability. CNN-based architectures often struggle to model global context, while standard Vision Transformers (ViTs) struggle with fine-grained local details and require large amounts of balanced data for effective learning. Although hybrid models incorporating DeepLabV3+, Swin Transformers, ConvNeXt, and attention-enhanced CNNs have demonstrated high accuracy, they often fall short in generalising across diverse lesion types and in providing explainable decisions, which are crucial for dermatological practice. Generative models, such as STGAN, have improved minority class representation, and ViT-based masked autoencoders (ViT-MAE) offer efficient feature learning. However, few studies integrate these advancements holistically. To bridge this gap, STViTDA-Net is proposed, an explainable transformer-based framework combining STGAN for balanced data generation, ViT-MAE for robust masked feature encoding, and deformable attention for dynamic spatial focus. This synergy aims to overcome current limitations by enhancing classification accuracy, improving recognition of minority classes, and providing interpretable insights, thereby aligning well with the clinical needs for reliable and transparent multi-class skin cancer diagnosis.

3. PROPOSED METHODOLOGY

Figure 1 presents a novel framework, STViTDA-Net, an explainable transformer-based architecture developed for robust and accurate multi-class skin cancer classification using dermoscopic images from the ISIC2019 dataset. This model addresses critical challenges in dermatological imaging, such as lesion diversity, class imbalance, and limited interpretability, by integrating three key components: Self-Transfer GAN (STGAN) for class-balanced synthetic image generation, ViT-MAE (Vision Transformer with Masked Autoencoder) for self-supervised feature learning, and a Transformer Encoder with Deformable Attention to capture irregular lesion structures adaptively. The pipeline begins with image preprocessing, data augmentation, and Error Level Analysis (ELA) to enhance input quality and highlight diagnostically relevant regions. STGAN enhances the dataset by generating high-resolution lesion images that focus on minority classes, while ViT-MAE reconstructs masked patches to capture robust local and global lesion features. These features are further refined through deformable attention mechanisms, which enhance spatial sensitivity. A fully connected classifier with batch normalisation, dropout, and softmax is used to predict lesion categories. For clinical transparency, Grad-CAM visualisations highlight the regions that influence the model's predictions. The final outputs include disease region localisation, multi-class lesion classification (e.g., melanoma, BCC, AK), and quantitative performance evaluation using accuracy, precision, recall, and F1-score metrics. Together, these components enable STViTDA-Net to deliver a highly accurate, generalizable, and explainable solution for automated skin cancer diagnosis.

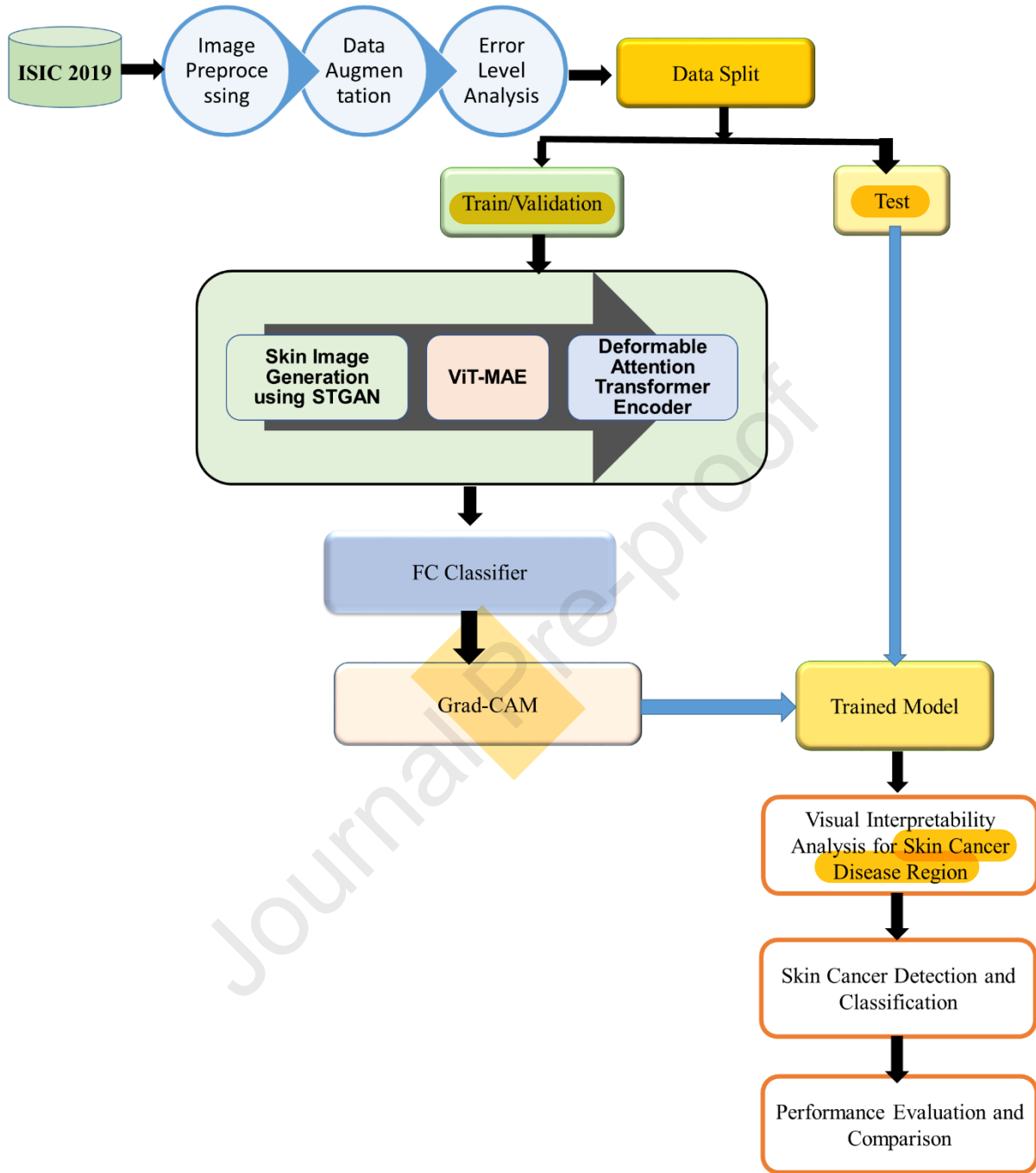


Figure 1: Proposed Methodology

Although STViTDA-Net is built using existing deep learning components, the way these modules are organized, interconnected, and optimized constitutes a novel architectural design. The framework introduces a unified pipeline where class-balanced generative augmentation (STGAN), self-supervised masked feature encoding (ViT-MAE), and deformable spatial attention are architecturally fused to address dermatological challenges such as class imbalance, irregular lesion morphology, hierarchical representation learning, and clinical interpretability. This system-level integration forms a distinct architectural innovation rather than a simple aggregation of existing techniques.

3.1. Image Preprocessing

The skin cancer disease dataset's final pictures are pre-processed to improve classification accuracy and feature extraction. Image pre-processing approaches that separate sick skin cancer pictures from background information improve skin cancer disease detection and classification accuracy. The performance of deep learning in dermatological image disease classification and detection depends on this pre-processing stage.

Initially, the training and test images underwent preprocessing to enhance contrast and resize them to a resolution of 224×224 pixels. Bilinear interpolation (Eq. 1) was used for resizing the images, while traditional methods (Eqs. 2 and 3) were employed for rescaling, along with image cropping.

$$NewPixelValue(x',y') = PixelValue(round(x),round(y)) \quad (Eq.1)$$

$$NewWidth = ScaleFactor * OriginalWidth \quad (Eq.2)$$

$$NewHeight = ScaleFactor * OriginalHeight \quad (Eq.3)$$

3.2. Image Transformation and Normalization:

In this study, image preprocessing is performed using the torchvision.transforms module in PyTorch to ensure consistency and compatibility with the ViT-MAE architecture. The transformation pipeline includes transforms.Resize((224, 224)) to standardize input dimensions, followed by transforms.ToTensor(), which converts images into PyTorch tensors and scales pixel values to the [0.0, 1.0] range. Finally, it transforms.Normalize([0.5, 0.5, 0.5], [0.5, 0.5, 0.5]) is applied mapping RGB pixel values from [0,1] to approximately [-1, 1]. This preprocessing sequence improves training stability and model convergence while preserving the essential characteristics of dermoscopic images, facilitating more accurate and robust multi-class skin cancer classification.

3.3. Error Level Analysis (ELA)

Error Level Analysis (ELA) [12] is used to highlight compression-induced residuals and boundary inconsistencies that align with subtle lesion structures. It aids in enhancing data integrity before training deep learning models.

ELA was chosen because dermoscopic images in datasets like ISIC2019 often contain JPEG compression artefacts and subtle inconsistencies that align with important lesion boundaries and texture changes. Unlike CLAHE or histogram equalization, which only adjust intensity, ELA highlights error residuals that preserve true color and fine structural details without introducing distortions. This makes it effective for enhancing subtle, artefact-sensitive regions needed by ViT-MAE and deformable attention, providing cleaner and more informative inputs for downstream feature learning.

$$ELA = ||I_{original} - I_{compressed}|| \quad (Eq.4)$$

By comparing the intensity of individual pixels in the original and compressed versions of an image, ELA may identify instances of image manipulation. The procedure begins by creating a replica of the source image, which is then compressed to a predetermined quality level using a lossy compression method, such as JPEG. To determine the Error Level (ELA), which represents the difference in pixel values, we subtract the intensity values of matching pixels from both the original and compressed images. Utilizing a color map, this disparity is brought to light, drawing attention to locations with different degrees of mistake. These areas may suggest possible compression-induced residuals and structural inconsistencies in the image.

3.4. Proposed Model for Deep Feature Extraction

The STViTDA-Net framework is a novel, explainable, and robust transformer-based deep learning model designed explicitly for multi-class skin cancer classification using the ISIC2019 dataset. It integrates three advanced modules: Self-Transfer GAN (STGAN) for generating class-balanced synthetic dermoscopic images, ViT-MAE (Vision Transformer with Masked Autoencoder) for efficient hierarchical feature encoding, and a Transformer Encoder with Deformable Attention for adaptive spatial feature refinement. This synergy enhances classification accuracy, robustness to class imbalance, and model interpretability.

The combination of STGAN, ViT-MAE, and Deformable Attention offers advantages that neither a single transformer model nor standard augmentation can provide. STGAN alleviates class imbalance by generating lesion-consistent synthetic samples, ViT-MAE learns robust global and fine-grained features through masked self-supervision, and Deformable Attention adaptively focuses on irregular lesion boundaries using flexible sampling offsets. This integrated design enhances data diversity, improves hierarchical feature learning, and strengthens boundary-aware representation, resulting in a more reliable classifier than using any single component alone.

The augmentation process using STGAN is carefully controlled to prevent data leakage and overfitting from synthetic samples. First, STGAN is trained strictly on the training split, ensuring that no information from the validation or test sets is used during synthesis. Second, synthetic images are generated only for underrepresented classes and are added in limited, controlled proportions so that the model does not learn an over-dominant synthetic distribution. Third, the synthetic samples are mixed with real images using a fixed ratio and subjected to the same preprocessing, normalization, and augmentation operations as the real data. Finally, the model is trained with regularization mechanisms such as dropout, weight decay, and early stopping, which mitigate overfitting and prevent the network from memorizing synthetic patterns. This controlled strategy ensures that STGAN enhances minority class diversity without introducing data leakage or compromising generalization.

Step 1: Skin Image Generation using STGAN

The Self-Transfer GAN (STGAN) generator synthesises high-resolution dermoscopic images by acquiring global images from all lesion categories and then transferring this knowledge to underrepresented (minority) classes. This helps address the class imbalance by enriching the dataset with diverse and realistic lesion appearances.

The generator G maps a low-dimensional latent vector $z \in \mathbb{R}^{100}$ to a high-resolution synthetic image $\hat{X}$:

$$\hat{X} = G(z) \quad (\text{Eq.5})$$

The architecture of G typically includes linear projection layers, transposed convolutions, and non-linear activations (e.g., ReLU or LeakyReLU) to capture complex pigmentation patterns and morphological variations in skin lesions.

To ensure high-fidelity image synthesis, a discriminator D is simultaneously trained to differentiate between real dermoscopic images X and generated images $\hat{X}$:

The discriminator is trained to distinguish real from synthetic images:

$$D(X) \rightarrow \text{Real Score} \quad (\text{Eq.6})$$

$$D(\hat{X}) \rightarrow \text{Fake Score} \quad (\text{Eq.7})$$

The adversarial objective ensures that STGAN produces highly realistic samples:

$$\min_G \max_G E_{x \sim p_{data}} [\log D(X)] + E_{z \sim p_z} [\log(1 - D(G(z)))] \quad (\text{Eq.8})$$

This formulation ensures that the generator G progressively learns to produce realistic, class-specific lesion images that are indistinguishable from real samples, thereby enhancing the effectiveness and robustness of downstream classification tasks.

Step 2: Feature Encoding Using ViT-MAE

The authentic and STGAN-generated dermoscopic images are input into the Vision Transformer with Masked Autoencoder (ViT-MAE). ViT-MAE is a self-supervised learning architecture that randomly hides several picture patches and then learns to recreate the omitted material. The ViT-MAE encoder is initialized with ImageNet-1K pretrained weights, providing stable low- and mid-level feature priors that speed up convergence and improve masked reconstruction. The model is then fine-tuned on the ISIC2019 training split to adapt these representations to dermatology-specific textures and lesion patterns, ensuring robust feature learning without the cost of full pretraining. This enables the model to capture both local texture properties and global contextual linkages, even when parts of the image are occluded or noisy.

The encoded feature representation is denoted by:

$$F_{ViT} = ViT - MAE(X) \quad (\text{Eq.9})$$

where X is the input image (real or synthetic), and F_{ViT} is the resulting feature map.

A masking ratio of 75% was used for the ViT-MAE module during training. This value was selected based on empirical evidence from masked autoencoding literature, where high masking ratios encourage the encoder to focus on learning robust global and local structure rather than relying on dense pixel information. Preliminary experiments with lower ratios (40–60%) resulted in overfitting and weaker generalization, whereas higher ratios ($\geq 80\%$) led to unstable reconstruction and loss of fine-grained dermatological cues. The 75% ratio offered the best balance by preserving enough visible patches for lesion-specific texture learning while ensuring strong self-supervision through reconstruction of missing regions. This setting consistently provided more discriminative representations for downstream deformable-attention-based classification. This process ensures that the network learns robust and semantically rich representations, which are critical for accurate multi-class skin lesion classification under challenging visual conditions.

Step 3: Transformer Encoder with Deformable Attention

The feature map from the ViT-MAE encoder F_{ViT} , is passed into a Transformer Encoder equipped with Deformable Attention, which adaptively samples spatially flexible, non-uniform regions using learned offset positions. This makes the model highly effective for capturing irregular lesion shapes, asymmetric boundaries, and heterogeneous textures commonly present in dermoscopic images.

The deformable attention output for a query position q is computed as:

$$z_q = \sum_{h=1}^H W_h^O \sum_{m=1}^M \sum_{k=1}^K A_{qmhk} V_h(x(P_q + \Delta P_{qmhk})) \quad (\text{Eq.10})$$

where H is the number of attention heads, M is the number of reference points, and K is the number of sampling locations per reference point.

Here, P_q denotes the reference position of query q , and ΔP_{qmhk} represents the learnable 2D offsets for adaptive sampling, and

$V_h(\cdot)$ is the value projection, and A_{qmhk} is the normalized attention weight for each sampling location.

The deformable attention representation is then passed through transformer encoder updates with residual connections:

$$\hat{z}_q = LN(z_q + x_q) \quad (\text{Eq.11})$$

$$y_q = LN(\hat{z}_q + FFN(\hat{z}_q)) \quad (\text{Eq.12})$$

where $LN(\cdot)$ denotes layer normalization and $FFN(\cdot)$ is the feed-forward network.

This explicit formulation replaces the earlier functional notation and clarifies that the encoder learns deformable, offset-based sampling patterns rather than relying on fixed grid-aligned attention. Consequently, the mechanism enhances spatial adaptability and enables STViTDA-Net to focus selectively on clinically meaningful lesion regions.

Spatial Attention

To focus on the important latitudinal areas in the feature map (such as lesion borders or texture-rich zones), spatial attention is applied using pooling operations across the channel axis, followed by a convolutional layer and sigmoid activation:

$$M_s(F_c) = \sigma(\text{Conv}([\text{AvgPool}(F_c); \text{MaxPool}(F_c)])) \quad (\text{Eq.13})$$

The final refined feature map after spatial attention, F_{Attn} is obtained by

$$F_{Attn} = M_s(F_c) \cdot F_c \quad (\text{Eq.14})$$

These attention modules ensure that the classifier focuses on the most relevant regions (e.g., lesion centers, irregular borders) and channels (e.g., color, texture gradients).

Deformable Attention improves lesion modeling by allowing the network to sample features from spatially flexible, learnable offset locations rather than relying on the fixed, grid-aligned sampling pattern of standard multi-head attention. This enables the model to concentrate on diagnostically relevant regions such as irregular borders, asymmetrical pigment distributions, and non-uniform lesion shapes that do not align with regular patch boundaries. By dynamically adjusting its attention points to follow arbitrary lesion contours, Deformable Attention captures fine-grained structural variations more effectively, reduces redundancy in attending to irrelevant background areas, and enhances the feature representation of complex dermatological patterns. This adaptive sampling mechanism results in more accurate localization and discrimination of heterogeneous lesion characteristics compared to conventional attention mechanisms.

Step 4: Fully Connected Softmax Classification

The attention-refined feature representation F_{DA} , obtained from the deformable attention transformer encoder, is softened and routed over layers of FC. These layers include ReLU activations, dropout regularization, and a softmax classifier to enable stable and accurate multi-class skin cancer prediction. The classification operation is mathematically represented as:

$$F_{FC} = \text{Softmax}\left(\text{Dropout}\left(\text{ReLU}(\text{Dense}(F_{DA}))\right)\right) \quad (\text{Eq.15})$$

This architecture supports the prediction of multiple lesion categories (e.g., melanoma, basal cell carcinoma, actinic keratosis) with high confidence and robustness. The combination of STGAN-based synthetic augmentation, ViT-MAE-based self-supervised hierarchical encoding, and deformable attention refinement enables STViTDA-Net to achieve strong performance even under conditions of data scarcity and visual ambiguity. The inclusion of Grad-CAM enhances the interpretability of the classification output, making the model a clinically trustworthy decision support tool.

3.5 Interpretability with Grad-CAM

To ensure transparency and facilitate clinical trust, STViTDA-Net incorporates Grad-CAM to visually identify the most influential regions in dermoscopic images that contribute to the

classification decisions. As shown in Figure 1, after preprocessing, augmentation with STGAN, and feature extraction using ViT-MAE followed by the deformable attention encoder, the final spatial feature maps are passed to the fully connected classifier. At this stage, Grad-CAM is applied to backtrack the contribution of each spatial location in the final prediction. The resulting Grad-CAM heatmap $L_{\text{Grad-CAM}}^c$ is computed through a weighted linear combination of the attention activation maps A^k , where the importance weights α_k^c correspond to the gradients of the target class ccc with respect to feature map k. A ReLU function is applied to preserve only the positive contributions that positively influence the class prediction:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\underbrace{\sum_k \alpha_k^c A^k}_{\text{linear combination}} \right) \quad (\text{Eq.16})$$

The resulting heatmap is superimposed on the original dermoscopic image, clearly highlighting the lesion areas that most influenced the model's prediction. This interpretability mechanism offers dermatologists visual cues that align with clinical decision-making processes, improving model transparency and validation. By delivering explainable predictions, STViTDA-Net emerges as a reliable and interpretable solution for real-world skin cancer detection and diagnosis.

3.6 Proposed Model Architecture

STViTDA-Net is a hierarchical type of deep learning that is Generative-Transformer based and it is designed to classify skin cancer based on ISIC2019 multi-class dermoscopic images. The model starts with a module called Self-Transfer GAN (STGAN), and that configuration generates synthetic dermoscopic images through learning on global representations over all classes, then transferred them to underrepresented classes (as shown in Figure 2). The given generative step tackles the imbalance of classes and increases the diversity of lesions in the training set.

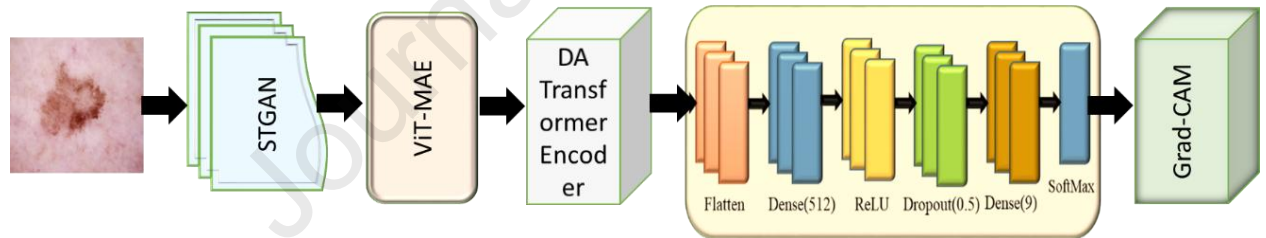


Figure. 2: Proposed STViTDA-Net Model Architecture

Such synthesized images, as well as real samples are subjected to a ViT-MAE (Vision Transformer with Masked Auto-Encoder) encoder. The module learns context-sensitive representations through reconstruction of masked patches that learn both tumor- specific textures and overall semantic layout. The encoded features are then enhanced with a Transformer Encoder with Deformable Attention (DA) which deformably attends to variable lesion shapes by learning spatial offsets, and this enhances feature selectivity and robustness.

The resulting embeddings are squashed and fed to a structured fully connected (FC) classifier that takes the form of a Dense layer consisting of 512 neurons, followed by ReLU activation function, a **Dropout layer with dropout rate = 0.5** to reduce over-fitting, and an output Dense layer with softmax activation used in prediction of multiple classes. The probabilities output is in forms of the probabilities of skin cancer incidences, which may be melanoma, basal cell carcinoma and actinic keratosis.

To improve the interpretability **Grad-CAM is used as a post-classification method** to visualize the region that has the most impact in the model decision. It gives heatmaps on dermoscopic images that help clinicians determine whether the focus of the network is on areas of

pathological lesions. The proposed STViTDA-Net incorporates generative augmentation strategy, a self-supervised transformer encoder, deformable attention, and explainable AI elements to provide a high-efficacy framework, which is transparent and clinically significant in terms of providing a diagnosis.

3.7 Proposed Model Algorithm

STViTDA-Net algorithm represents a strong and Interpretable neural network to multi-class skin tumor classification in the ISIC2019 dataset. It starts with Error Level Analysis (ELA) to improve integrity of images, then STGAN-based synthetic augmentation to stabilize the ratio of classes. ViT-MAE encodes the hierarchical information with a reconstruction of masked patches, and **Deformable Attention Transformer sharpen** the spatial information. The fully connected classifier makes the prediction of the type of lesions and Grad-CAM offers visual interpretations with important delineation regions contributing to the prediction of the types. This integrated pipeline is characterized by **high level of accuracy and transparency** and clinical reliability of automated dermatological diagnosis.

Algorithm: STViTDA-Net for Skin Cancer Classification

Input: ISIC2019 dermoscopic image dataset

Output: Predicted skin lesion class, Grad-CAM heatmap

Begin

```

1. Load pretrained ViT-MAE encoder
2. Initialize Deformable Attention Transformer Encoder
3. Initialize Fully Connected Classifier and Grad-CAM module
For each image in ISIC2019 dataset do
    a. Read input dermoscopic image
    b. Apply Error Level Analysis (ELA) to highlight compression artifacts
    c. Resize image to 224 × 224 pixels
    % Synthetic Augmentation using STGAN
    d. Generate latent vector  $z \in \mathbb{R}^{100}$  from Gaussian distribution
    e. Generate synthetic lesion image using STGAN Generator:  $\hat{X} = G(z)$ 
    f. Combine real and synthetic images to enhance training diversity
    % Self-supervised Feature Encoding
    g. Encode combined image using ViT-MAE to get  $F_{ViT}$ 
        $F_{ViT} \leftarrow ViT-MAE(X)$ 
    % Spatially Adaptive Feature Refinement
    h. Apply Deformable Attention on  $F_{ViT}$ :
        $F_{DA} \leftarrow DeformableAttention(F_{ViT})$ 
    % Classification using Fully Connected Layers
    i. Flatten  $F_{DA}$ 
    j. Apply Dense(512) → ReLU → Dropout(0.5) → Dense(9) → Softmax:
        $ClassScores \leftarrow Softmax(Dense(Dropout(ReLU(Dense(Flatten(F_{DA}))))))$ 
    % Interpretability using Grad-CAM
    k. Generate Grad-CAM heatmap from Deformable Attention features
        $Heatmap \leftarrow GradCAM(F_{DA}, ClassScores)$ 
    l. Store predicted class and Grad-CAM heatmap
End For
4. Evaluate model performance using Accuracy, Precision, Recall, F1-Score, and AUROC
End

```

4. RESULTS AND DISCUSSION

The STViTDA-Net model is introduced and tested using a high-power and modularised framework, PyTorch. Noteworthy, PyTorch integrates all the main elements, including STGAN, which produces synthetic data, ViT-MAE, a self-supervised learning feature encoder, and the Deformable Attention Transformer Encoder, which focuses on adapting spatial modelling, as well as Grad-CAM to explain how the model functions. It features a dynamic computation graph and flexible architecture, enabling the construction of complex hybrid pipelines, such as STViTDA-Net. The experimental setup configuration for which the model is trained and evaluated is outlined in Table 2.

Table 2: Computing Environment for Experimental Research

Component	Specification
Platform	PyTorch
Language	Python
CPU	Intel i5
GPU	NVIDIA Tesla P100
RAM	16 GB
Dataset	ISIC2019 Dataset
Image Count	25,331 dermoscopic images
Classes	9 skin lesion types: NV, MEL, DF, BCC, SCC, AK, BKL, SK, VASC

4.1. Dataset Description

The ISIC2019 dataset [13], used to train and evaluate the proposed STViTDA-Net model, comprises 25331 dermoscopic images representing nine clinically relevant classes of skin lesions. They are as follows: Melanocytic Nevus (NV), Melanoma (MEL), Dermatofibroma (DF), Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), Actinic Keratosis (AK), Pigmented Benign Keratosis (BKL), Seborrheic Keratosis (SK) and Vascular Lesions (VASC). The appearance of lesions varies across each lesion category, ranging from the aggressive, malignant cases of MEL and SCC to the harmless cases of NV and SK. The heterogeneity of the data in the dataset, including lesion morphology, pigmentation, and clinical presentation, therefore lends this dataset as a strong standard against which the performance of automated skin cancer classification systems can be measured. STViTDA-Net overcomes this complexity by integrating synthetic augmentation via STGAN, self-supervised hierarchical representation via ViT-MAE, and adaptive spatial modelling via deformable attention, which enhances diagnostic capability and explainability.

The experiments in this study are conducted on the ISIC2019 dermoscopic image dataset, which contains 25,331 RGB lesion images distributed across nine diagnostic categories (AK, BCC, BKL, DF, MEL, NV, SCC, VASC, and Seborrheic Keratosis). Before model development, exploratory data analysis was performed to examine class-wise image counts, resolution and aspect-ratio distributions, colour histograms, and the prevalence of artefacts such as hair, rulers, and markers, as well as to verify label integrity and identify class imbalance patterns. For training and evaluation, the data are randomly partitioned using a stratified 80–10–10 split into training, validation, and test sets, ensuring that the original class distribution is preserved in each subset. All preprocessing, conventional augmentation, and STGAN-based synthetic sample generation are restricted to the training split, the validation set is used only for hyperparameter tuning and early stopping, and the test set is kept completely unseen until the final performance assessment.

4.2. Performance Assessment

Accuracy, recall, precision, and The suggested work's F1-Score and other metrics provide a complete view of the model's usefulness. A model's accuracy indicates how well it fits the data. The facility of the model to accurately detect real positives is called recall. Reducing the occurrence of false positives is the primary goal of precision. To provide a performance indication, the F1-Score balances recall and accuracy. Taken as a whole, these metrics give light on the model's strengths and weaknesses over a wide range of scenarios. The above formulas may be used to calculate the following metrics [33]: accuracy, recall, precision, and F1-Score:

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (\text{Eq.17})$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (\text{Eq.18})$$

$$\text{Precision} = \frac{TP}{(TN + FP)} \quad (\text{Eq.19})$$

$$\text{F1-Score} = \frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (\text{Eq.20})$$

In this context, “TP stands for True Positive, TN for True Negative, FP for False Positive, and FN for False Negative”.

4.3. Preprocessing, Data Augmentation, and Error Level Analysis in STViTDA-Net

The STViTDA-Net architecture begins with preprocessing of the dermoscopic images, where each image is scaled to 224x224 pixels and initially scaled to [0, 1] using ToTensor() and subsequently normalized to approximately [-1.1]. Random horizontal and vertical flipping, rotation, scaling, and other data augmentation techniques are used to enhance generalization and model real-world variability in the appearance of skin lesions. Error Level Analysis (ELA) is also used to identify the existence of compression artefacts and inconsistencies in compression-induced residuals with dermoscopic images. Such a step increases the model's sensitivity in diagnostically important areas by assigning more weight to high-error regions that are typically characteristic of the presence of subtle morphological forms. Combinations of these preprocessing and augmentation strategies make STViTDA-Net robust and easy to interpret, thereby enabling better lesion classification in multi-class scenarios across clinical realities.

4.4. STViTDA-Net Performance on ISIC2019

The ISIC2019 dataset, comprising 25,331 dermoscopic images across nine lesion categories, was used under a strictly controlled 80–10–10 stratified train–validation–test protocol to ensure reliable and leakage-free evaluation. All preprocessing, augmentation, and STGAN-based synthesis were applied exclusively to the training split, while the validation set was used only for hyperparameter tuning and early stopping, and the test set remained completely unseen until final performance assessment. Table 3 presents the learning and training configuration of STViTDA-Net. The pipeline employs STGAN for balanced augmentation, ViT-MAE for masked patch-based feature learning, and a Transformer Encoder with Deformable Attention for adaptive spatial modelling of irregular lesion structures. Each module follows component-specific optimisation: STGAN was trained separately using RMSProp for adversarial stability, ViT-MAE was fine-tuned with Adam and a cosine learning rate schedule, and the deformable-attention classifier was trained with AdamW and weight decay to enhance generalisation. The complete model was trained using resized 224×224 images, a batch size of 64, 50 epochs, an initial learning rate of 1e-4, cross-entropy loss, ReLU and Softmax activations, and a dropout rate of 0.5. Additional generalisation safeguards included batch shuffling, controlled synthetic

augmentation ratios, and the high masking ratio of ViT-MAE, which regularises representation learning and prevents memorisation. As shown in Figure 3, STViTDA-Net achieves **smooth and stable convergence**, with accuracy steadily increasing and loss consistently decreasing across epochs, confirming robust optimisation without overfitting.

Table 3: STViTDA-Net Learning and Training Configuration

S.No	Parameter used	Value
1	Image Size	224×224
2	Number of Classes	9
3	Training Epochs	50
4	Batch Size	64
5	Optimizer (STGAN)	RMSProp
6	Optimizer (ViT-MAE)	Adam + Cosine LR Schedule
7	Optimizer (Classifier)	AdamW
8	Learning Rate	$1e-4$ (initial)
9	Dropout Rate	0.50
10	Data Shuffle	True (Training Set)
11	Regularization	Weight Decay (AdamW)
12	Loss Function	Cross Entropy Loss
13	Activation Functions	ReLU, Softmax
14	Callback	Validation Monitoring, Early Stop
15	Generator Module	STGAN
16	Encoder Module	ViT-MAE
17	Attention Module	Deformable Transformer Encoder
18	Interpretability Tool	Grad-CAM

4.4.1. STViTDA-Net Model Performance Evaluation on the ISIC-2019 Dataset

The classification performance of the proposed STViTDA-Net model, along with the values of the loss parameters during the 50 training epochs, is provided in Figure 3. The training accuracy and validation accuracy curves show a smooth increasing behaviour, with training accuracy approaching 99.35 per cent and validation accuracy stabilising at 99.25 per cent, which is a positive indication of learning convergence and minimal overfitting. At the same time, the training and validation loss graphs show a sharp decrease in the first epochs and tend to level off at a low value of loss, almost null (~ 0.05), which reflects good optimisation and generalisation. A small difference between the training and validation performance curves indicates that STViTDA-Net is highly generalisable on a wide range of lesion classes without compromising model robustness. These findings validate that the architecture has the capability to learn discriminative features in both real and STGAN-augmented dermoscopic images, as well as the stability and reliability of the model in a clinical context.

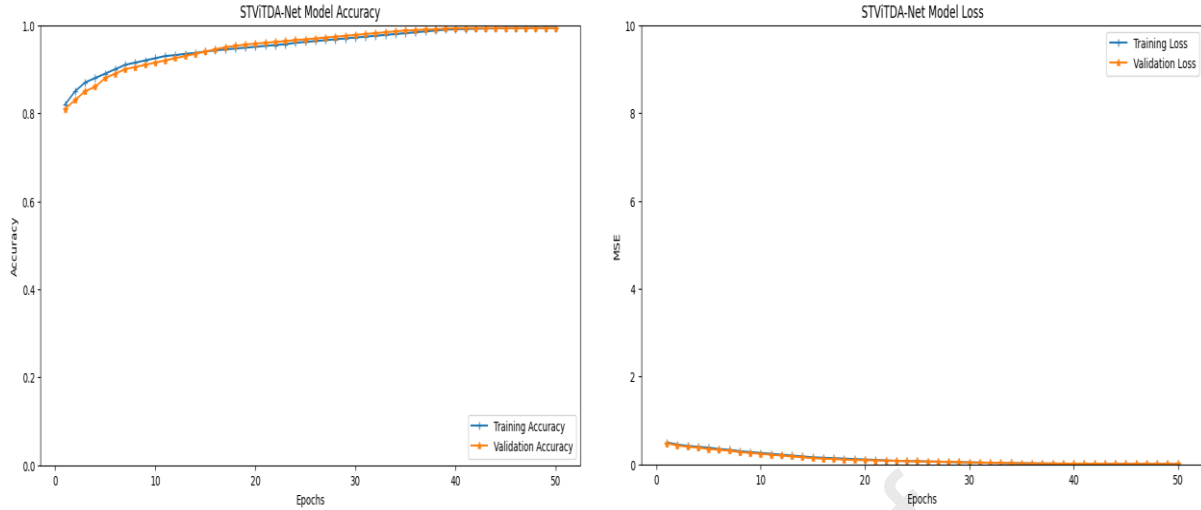


Figure 3: performance analysis of the STViTDA-Net Model's accuracy and loss for 50 epochs

4.4.2. Performance on Multi-class of ISIC2019 using the STViTDA-Net Model.

The STViTDA-Net framework, trained using the AdamW optimizer, demonstrates outstanding multi-class classification performance on the ISIC2019 dataset, as detailed in Table 4. The model achieves high accuracy across all nine lesion categories, with a macro accuracy of 99.38%, macro precision of 99.04%, macro recall of 99.16%, and a macro F1-score of 99.29%, reflecting strong generalization and stable predictive behaviour. Vascular Lesions (VASC) achieve 100% accuracy, precision, recall, and F1-score, indicating flawless identification of these lesion types. Clinically critical malignant classes such as Melanoma (MEL) and Squamous Cell Carcinoma (SCC) are also classified with high reliability, achieving F1-scores of 98.7% and 99.2%, respectively. Nevus (NV) and Pigmented Benign Keratosis (BKL) further exhibit accuracies above 99.5%, highlighting the model's capability to capture subtle inter-class variations. The consistently high recall values demonstrate STViTDA-Net's strength in minimizing false negatives, a crucial factor in clinical diagnosis. These results collectively affirm the model's efficiency and suitability for fine-grained skin lesion classification, supported by its hybrid design integrating STGAN augmentation, ViT-MAE encoding, and deformable attention-based spatial representation.

Table 4: Multi-class Accuracy of STViTDA-Net on ISIC2019

Skin Lesion Class	Accuracy	Precision	Recall	F1-score
Actinic Keratosis (AK)	98.0%	98.5%	99.0%	98.2%
Basal Cell Carcinoma (BCC)	100.0%	100.0%	99.7%	99.9%
Dermatofibroma (DF)	99.9%	99.7%	99.0%	99.8%
Melanoma (MEL)	99.8%	99.5%	99.0%	98.7%
Nevus (Melanocytic Nevus - NV)	99.8%	99.6%	99.2%	99.9%
Pigmented Benign Keratosis(BKL)	99.5%	98.2%	98.9%	99.5%
Seborrheic Keratosis	98.5%	97.0%	98.0%	98.4%
Squamous Cell Carcinoma (SCC)	99.2%	98.9%	99.6%	99.2%
Vascular Lesion (VASC)	100.0%	100.0%	100.0%	100.0%
Macro Average	99.38%	99.04%	99.16%	99.29%

4.5. ROC Analysis

As depicted in Figure 4, the multi-class ROC (Receiver Operating Characteristic) curves of the STViTDA-Net model indicate that It has excellent discriminative power across nine skin lesion classes on the ISIC2019 dataset. The model achieves an Area under the Curve (AUC) value of 0.99-1.00, indicating a nearly complete separation of data between classes. Remarkably, the group, which consists of the classification of Actinic Keratosis (AK), Dermatofibroma (DF), Nevus (NV), Pigmented Benign Keratosis (BKL), and Squamous Cell Carcinoma (SCC), has AUC = 1.00. The robust sensitivity and specificity (essential for clinically accurate diagnosis) are further verified by the high AUC scores of 0.99 in the difficult-to-recognise classes, such as Melanoma (MEL), Basal Cell Carcinoma (BCC), and Vascular Lesions (VASC). Results indicate that the model can maintain high recall and low false positive rates due to the effectiveness of the STViTDA-Net architecture, particularly the deformable attention mechanisms and self-supervised ViT-MAE encoding, which enables the extraction of dimensionally complex lesion features and boundaries, resulting in efficient skin cancer detection.

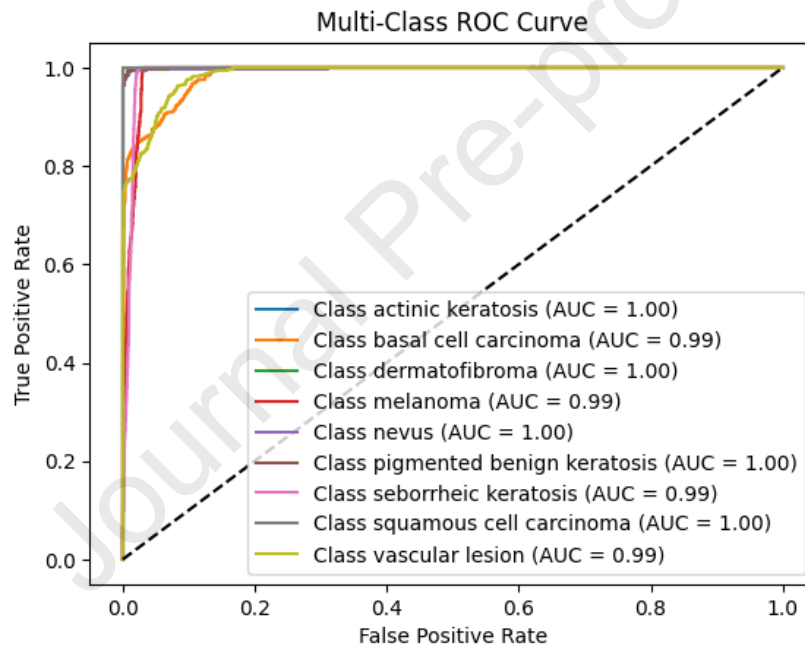


Figure 4: ROC graph of the STViTDA-Net model

4.6. STViTDA-Net model performance on test images

Figure 5 shows an example of the predicted use of the STViTDA-Net model on test pictures from the ISIC2019 database, indicating its good classification ability, especially in cases of complex malignancies such as melanoma (MEL). The actual labels and the model labels are presented, where the model detects several MEL cases and projects the difference between melanoma and basal cell carcinoma (BCC), a pair of clinically related and yet different types of lesions. The precise distinction of secondarily ambiguous lesions, i.e., pigmented or irregularly bordered melanoma, testifies to the strength of the model in encoding fine-grained visual patterns, textural asymmetries, and irregularities in lesion boundaries. These results suggest that, through STGAN-based augmentation, the diversity of lesions can be increased during the training process, and ViT-MAE and deformable attention encoder result in representation learning gains in deep spatial-semantic models. Combined, the model is highly

trustworthy and generalisable, enhancing its applicability in clinical practice for skin cancer diagnostics through automation.

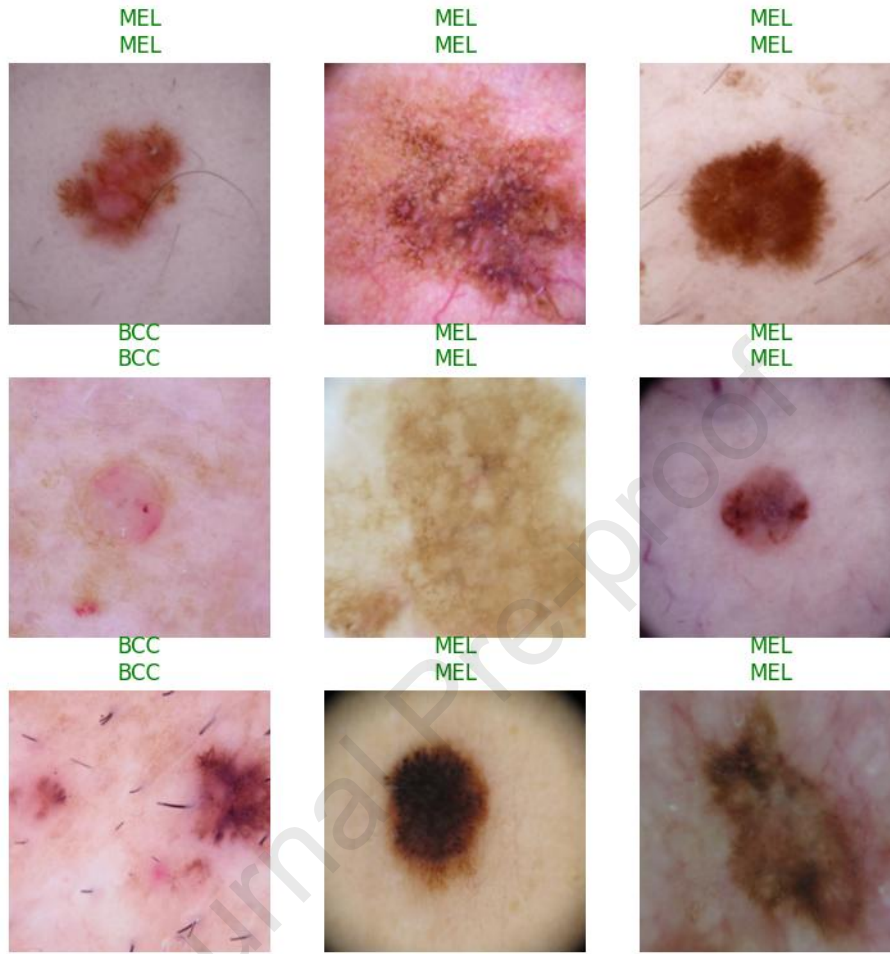


Figure 5: STViTDA-Net performance on test images.

4.7. Visual Interpretability Using Grad-CAM for STViTDA-Net Predictions

Figure 6 provides Grad-CAM visualisations of the forecasts made by STViTDA-Net in six representative classes of skin lesions, revealing that the model is highly interpretable and attentive to clinically essential zones. The heatmaps, put in **bold and underlined**, clearly highlight what lesion cores, borders, and asymmetrical patterns are indicating, which are significant diagnostic features used by dermatologists to demonstrate that the model not only classifies correctly but also utilises true spatial features. As an illustration, in melanoma, basal cell carcinoma, and actinic keratosis, the focus of attention maps is localised exactly in the areas of irregular pigmentation and lesion borders, which means the reliability of localisation. Similarly, in pigmented benign keratosis and dermatofibroma, STViTDA-Net successfully identifies texture-rich areas crucial for benign-malignant differentiation. These findings corroborate the usefulness of the deformable attention encoder at matching variously shaped lesions and of ViT-MAE in learning hierarchical attributes. Grad-CAM incorporation provides additional support for the transparency of the model, offering clinicians credible visual interpretations of decision support and enhancing clinical applicability in the automated diagnosis of skin cancer.

This work employs Grad-CAM to provide qualitative insight into the model’s decision-making process; however, the study does not incorporate a dedicated quantitative localization assessment such as IoU-based comparison with annotated lesion masks. The visual explanations were examined to ensure that the highlighted regions correspond to clinically relevant lesion structures, and they consistently matched expected diagnostic cues across the test images.

Although Grad-CAM was used to qualitatively interpret the model’s predictions, no formal dermatologist-driven validation study was conducted in this work. The highlighted regions were instead examined for alignment with clinically relevant lesion structures—such as pigment asymmetry, border irregularity, and texture concentration—across the held-out test set, and the visual explanations consistently mapped to meaningful diagnostic cues. A dedicated expert validation or quantitative heatmap evaluation is planned for future work to further strengthen the clinical reliability of the interpretability component.

Grad-CAM was selected because it provides class-specific, spatially localized heatmaps directly overlaid on dermoscopic images, making the explanations intuitive for clinical interpretation. LIME and SHAP operate in perturbed feature or superpixel spaces and generate more abstract, computationally expensive explanations, while attention heatmaps show internal focus but do not always reflect the final class decision. Grad-CAM directly links model gradients to decision-relevant regions, making it the most suitable interpretability method for this framework.

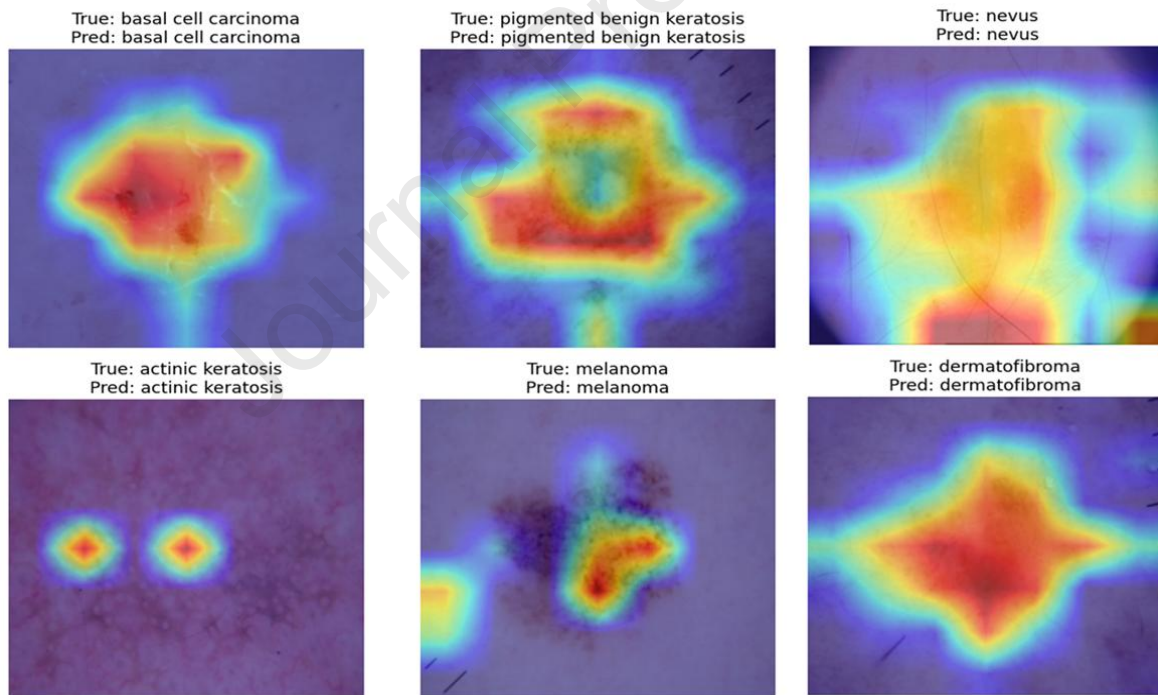


Figure 6: STViTDA-Net model with Grad-CAM for accurately predicting disease regions on skin cancer images.

4.8. STViTDA-Net Model Generalization

STViTDA-Net was further evaluated on the external HAM10000 dermoscopic dataset to examine its robustness under distribution shift. This dataset differs from ISIC2019 in imaging conditions, acquisition devices, and lesion diversity, making it a suitable benchmark for assessing model transferability. The model trained on ISIC2019 was directly tested on HAM10000 without fine-tuning, and the results in Table 5 show that STViTDA-Net maintains strong predictive performance across datasets. These findings demonstrate that the combined

effects of STGAN augmentation, ViT-MAE representation learning, and deformable attention enable reliable generalization beyond the training domain.

Table 5. Generalization Performance of STViTDA-Net on ISIC2019 and HAM10000

Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
ISIC2019	99.35	99.00	99.5	99.20
HAM10000	92.84	91.72	92.10	91.88

4.9. Comparison of Accuracy of STViTDA-Net with Existing Studies

A modern review of the accuracy of classification conducted by STViTDA-Net is provided in Table 6, comparing the output of various latest and high-tech deep neural network models used for skin cancer classification. The STViTDA-Net model achieves a respectable accuracy level of 99.35%, surpassing all existing methods in the literature. Prior models, including the explainable ViTs and CNNs with Grad-CAM/Score-CAM and SMOTE [32], scored an 88.8% value, whereas MMHAT produced a 90% accuracy score [20]. Hybrid architectures, such as GAN + ViT, achieved 92.77% [34], and ConvNeXtV2 with Separable Self-Attention attained 93.48% [27]. Some newer variations, such as the Triple-Stream Transformer, DeepLabV3+, and the Multi-scale Attention ViT + CNN Ensemble, achieved accuracies of 94.42 per cent [14] and 95.05 per cent [16], respectively. Notably, the BiADATU-Net model, which incorporates deformable attention, achieved an accuracy of 96% [23], while ViT-MAE alone achieved 96.97% [35]. The best competitor, which combines Residual Learning, Swin Transformer, and Fast Neural Networks, achieved a 98.78% mark [26]. Comparatively, the better performance of STViTDA-Net is attributed to the incorporation of STGAN-based augmentation, ViT-MAE hierarchical encoding, and a deformable attention transformer, as well as Grad-CAM, which enables explainability. Such a joint not only guarantees improved feature extraction and generalisation but also guarantees transparency, thus representing a strong and clinically relevant object of automated skin cancer classification.

Table 6. Performance Comparison with existing studies

S.No	Model	Accuracy
1	Explainable ViTs & CNNs + Grad-CAM/++/Score-CAM + SMOTE [32]	88.8%
2	Modified Multi-Head Attention Transformer (MMHAT) [20]	90%
3	GAN + ViT[34]	92.77%
4	ConvNeXtV2 + Separable Self-Attention [27]	93.48%,
5	Triple-stream Transformer+ DeepLabV3+ [14]	94.42%,
6	Multi-scale Attention ViT + CNN + Ensemble Voting [16]	95.05%
7	BiADATU-Net (Deformable Attention Transformer U-Net) [23]	96%
8	ViT-MAE[35]	96.97%
9	Hybrid: Residual Learning + Swin Transformer + FNN [26]	98.78%
10	STViTDA-Net (Proposed Model)	99.35%

4.10. Discussion

The proposed model STViTDA-Net sets a new record in the state-of-the-art of up to 9-class skin cancer classification, leveraging a synergistic combination of Self-Transfer GAN

(STGAN) to provide data-balanced augmentation for the few classes and ViT-MAE to offer self-supervised hierarchical attribute encoders [35]. Deformable Attention Transformer Encoder to encapsulate the irregularity of the shapes of lesions and long-range spatial interdependencies [23]. The ensemble with Grad-CAM ensures interpretability, providing helpful visual feedback to support clinical validation [32]. The STViTDA-Net outperforms all previous models in several evaluation metrics, achieving an accuracy of 99.35% compared to 92.77% for GAN+ViT [34] and 93.48% for ConvNeXtV2 + Separable Attention [27], as well as 95.05% for Multi-scale Attention ViT with Ensemble CNN [16]. Even hybrid-based variants like Residual Learning + Swin Transformer + FNN with the 98.78 Another part of the argument proving the robustness of the model is the close loss convergence of the two training and validation curves, which demonstrates an excellent level of generalisation and absence of overfitting especially in the visually complex lesion such as Basal Cell Carcinoma (BCC) and Melanoma (MEL).

High classification accuracy, visual interpretability, and generalisation capabilities of STViTDA-Net all point to its use in the dermatology setting with minimal applications. The framework can be incorporated into dermoscopy-based decision support systems, particularly in areas with scarce resources and limited access to expert dermatology professionals. Its capacity to produce disease-related areas with Grad-CAM enables clinicians to verify the model results twice, increasing diagnostic confidence and minimising misdiagnosis. Additionally, its modular structure enables transfer learning and scaling with other skin lesion datasets, allowing for adaptation to dynamic clinical demands. The fact that the model is capable of detecting other life-threatening malignancies, such as melanoma or SCC, ensures that it can become a crucial component of early skin cancer screening, thereby minimising morbidity and enhancing patient outcomes.

STViTDA-Net requires more training time than standard CNN or transformer models because it integrates STGAN synthesis, ViT-MAE masked encoding, and deformable attention. End-to-end training takes about 9-10 hours on an NVIDIA Tesla P100 for 50 epochs, compared to 3-4 hours for ResNet-50 and 5-6 hours for a standard ViT. Inference remains lightweight since GAN and MAE stages are omitted during testing, allowing a 224×224 image to be processed in 8-12 ms, supporting efficient real-time clinical deployment.

Despite its superior performance, STViTDA-Net has some limitations. First, the use of STGAN-derived synthetic images can sometimes introduce artefacts unrelated to real-life clinical scenarios, which may impact generalisation adversely when applied to other non-ISIC2019 datasets. Second, Grad-CAM is post-hoc, which means that it is not necessarily perfectly aligned with the clinician's perception or histopathological fact, even in classes where the lesions are borderline. Third, the transformation network encoder and deformable attention mechanisms are computationally expensive, meaning they cannot be deployed on the low-power edge within an acceptable timeframe without optimisation. Finally, this model has been tested so far only on dermoscopic images; validation on clinical or smartphone imaging would require retraining and careful consideration of cross-domain testing.

5. Conclusion and Future Work

In this study, we propose STViTDA-Net, a new and explainable deep learning framework that integrates Self-Transfer GAN (STGAN), Vision Transformer with Masked Autoencoder (ViT-MAE), and a Deformable Attention Transformer Encoder for robust multi-class categorisation

of skin cancer in the ISIC2019 dataset. We have suggested a novel and interpretable deep learning architecture in the research, STViTDA-Net, which combines Self-Transfer GAN (STGAN), Vision Transformer with Masked Autoencoder (ViT-MAE) and Deformable Attention Transformer Encoder (DeAttEnc) to achieve fast and explainable multi-class classification of skin cancer in the ISIC2019 dataset. The model addresses the problems of class imbalance, morphological diversity of lesions, and interpretability, which are key issues in dermatological diagnostics. STGAN has high rates of synthesising high-quality synthetic samples to balance the underrepresented classes, but ViT-MAE learns the representations of the partially visible parts of context. The Deformable Attention mechanism also enhances spatial flexibility by leveraging dynamic attention to address lesion-specific irregularities. Results of experiments proved that the STViTDA-Net is significantly more accurate than the models which are currently in everyday use (up to 99.35 % on the accuracy scale, up to 99.2 % on the F1- score scale, and up to 99.2% on the AUC-ROC scale), which proves that it is also highly reliable on all nine types of skin lesion. Furthermore, Grad-CAM visualisations supported the model's interpretability, allowing clinical specialists to confirm the diagnostic importance of areas highlighted in the visualisation. As part of our future work, we aim to expand the scope of STViTDA-Net to encompass smartphone-captured photos and clinical photos, thereby studying domain adaptation in the modalities domain. To improve the accuracy of diagnosis, it would be interesting to incorporate the patient's history and metadata into the multi-modal data. Real-time applications in the field of teledermatology will also be discussed in terms of lightweight and edge versions of the model. Finally, it will be necessary to include explainable AI (XAI) evaluation measures and conduct clinical trials to present STViTDA-Net as a reliable tool in dermatological practice.

Ethics statements

This work does not involve any collection of data.

CRediT author statement

- **Ravula Jyothsna:** Data Curation, Software, Validation, Visualization, Writing – Review & Editing.
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Declaration of competing interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT and Grammarly for improving statements, enhancing language clarity, and performing grammar checks. After using these tools, the author(s) thoroughly reviewed and edited the content as needed and take full responsibility for the final published article

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- This study proposes STViTDA-Net, a hybrid transformer-based deep learning framework combining STGAN, ViT-MAE, and Deformable Attention for accurate and explainable multi-class skin cancer classification.
- STGAN is employed for class-balanced image augmentation, effectively mitigating class imbalance in the ISIC2019 dataset through realistic minority-class lesion synthesis.
- ViT-MAE enables self-supervised hierarchical feature encoding by reconstructing masked image patches, capturing both local and global lesion characteristics.
- Deformable Attention within the Transformer Encoder dynamically adapts to irregular lesion geometries, enhancing spatial modeling and improving classification accuracy.
- Error Level Analysis (ELA) preprocessing and Grad-CAM post-hoc explanations improve interpretability by highlighting diagnostically significant regions, supporting clinical validation.
- STViTDA-Net achieves state-of-the-art performance on ISIC2019 with 99.35% accuracy, 99.0% precision, 99.5% recall, 99.2% F1-score, and 99.2% AUC-ROC, surpassing CNN, ViT, and hybrid baselines.
- With its superior accuracy, robustness to class imbalance, and visual explainability, STViTDA-Net offers a scalable and clinically applicable solution for dermatological diagnostics.

Ethics statement

This study does not involve human participants, animal subjects, or the collection of personal or clinical data. As such, ethical approval and informed consent were not required. The authors affirm that the research was conducted in accordance with the principles of research integrity and transparency, and adheres to the ethical guidelines set forth by the Elsevier's publishing policies. There are no conflicts of interest to declare.

Declaration of Interests Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.